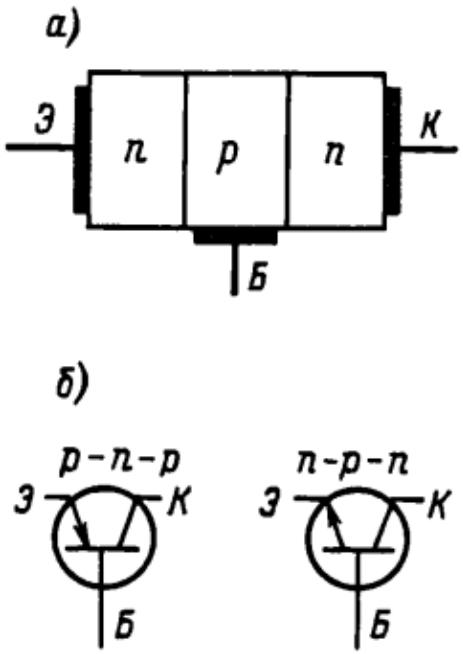
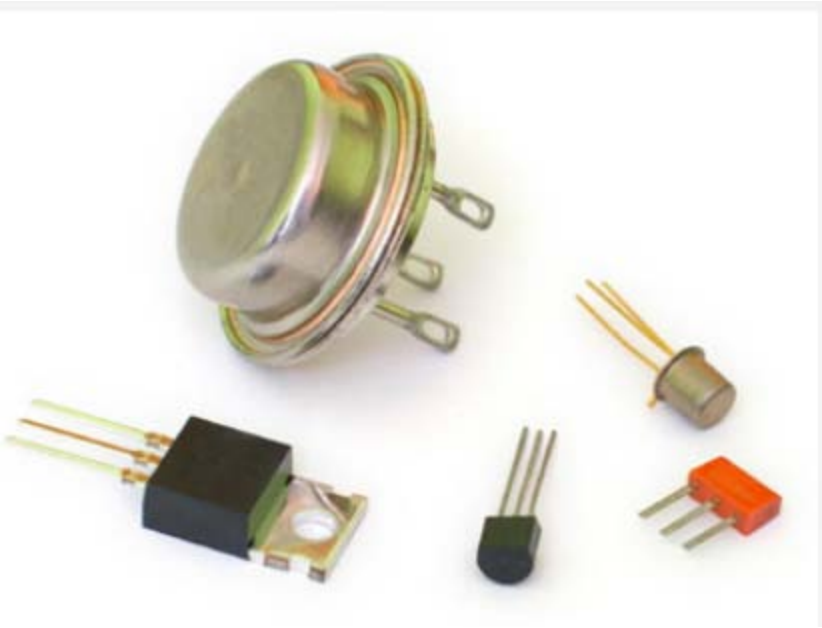
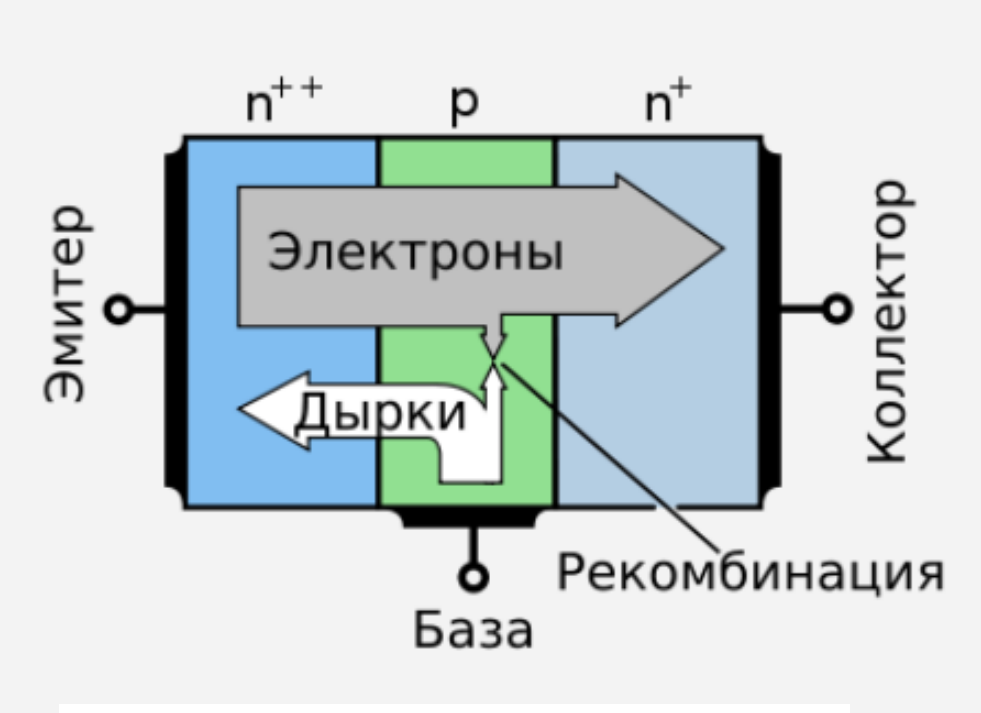
Биполярные транзисторы



Принцип устройства (*а*) и условное графическое обозначение (*б*) плоскостного транзистора

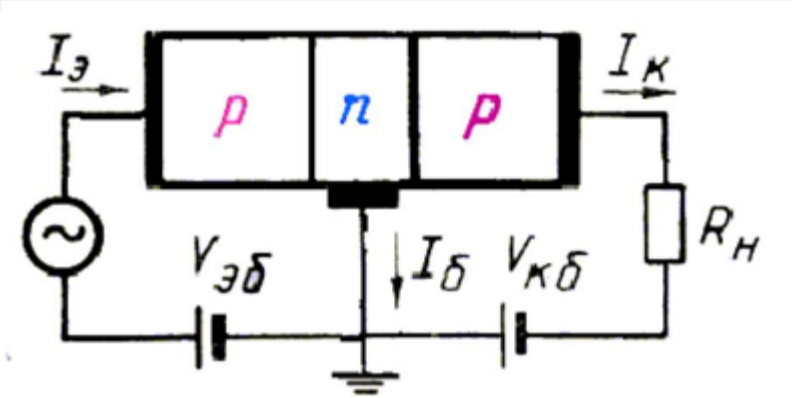


Биполярные транзисторы



$$I_э = I_б + I_к$$

$$\alpha = \left(\frac{\partial I_к}{\partial I_э}\right)_{U_к = const},$$



$$I_к = \alpha\, I_э,$$

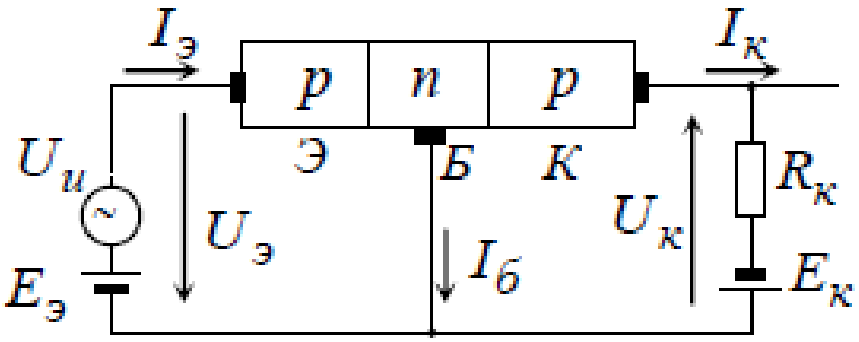
$$I_б = I_э - I_к = (1 - \alpha)\, I_э.$$

Биполярные транзисторы

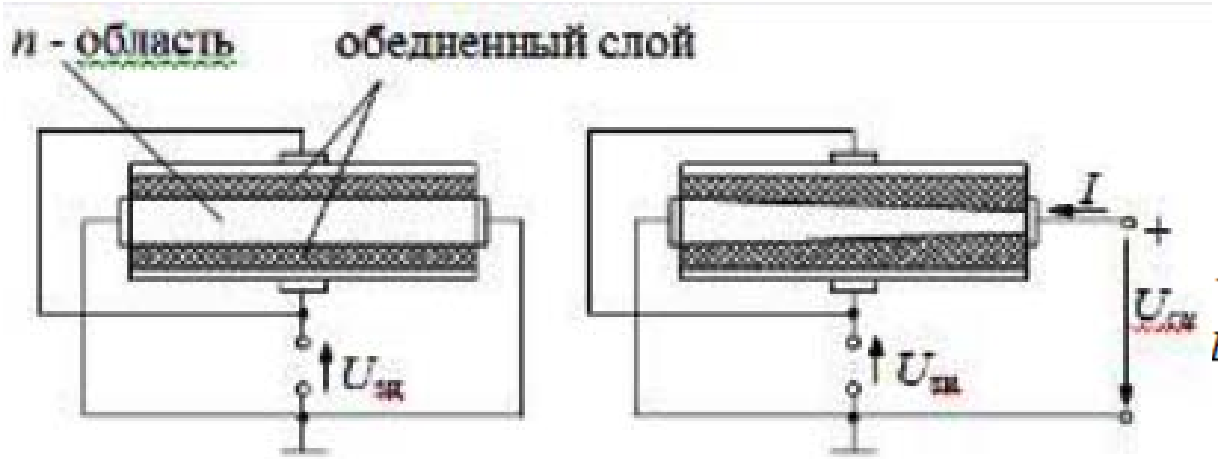
$$U_{вх} = I_э \cdot R_{вх}, \quad U_{вых} = I_к \cdot R_к = \alpha I_э \cdot R_к.$$

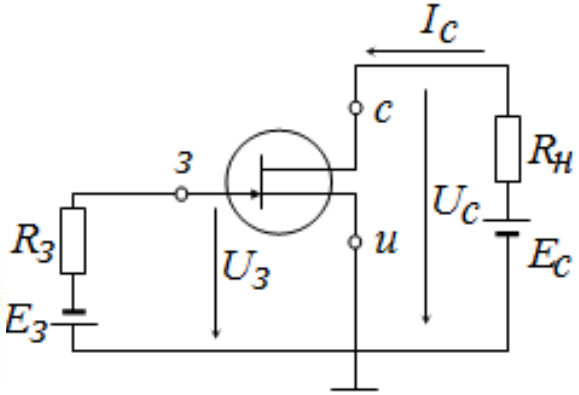
$$K_u = \frac{U_{вых}}{U_{вх}} = \frac{I_к R_к}{I_э R_{вх}} = \frac{\alpha R_к}{R_{вх}} >> 1$$

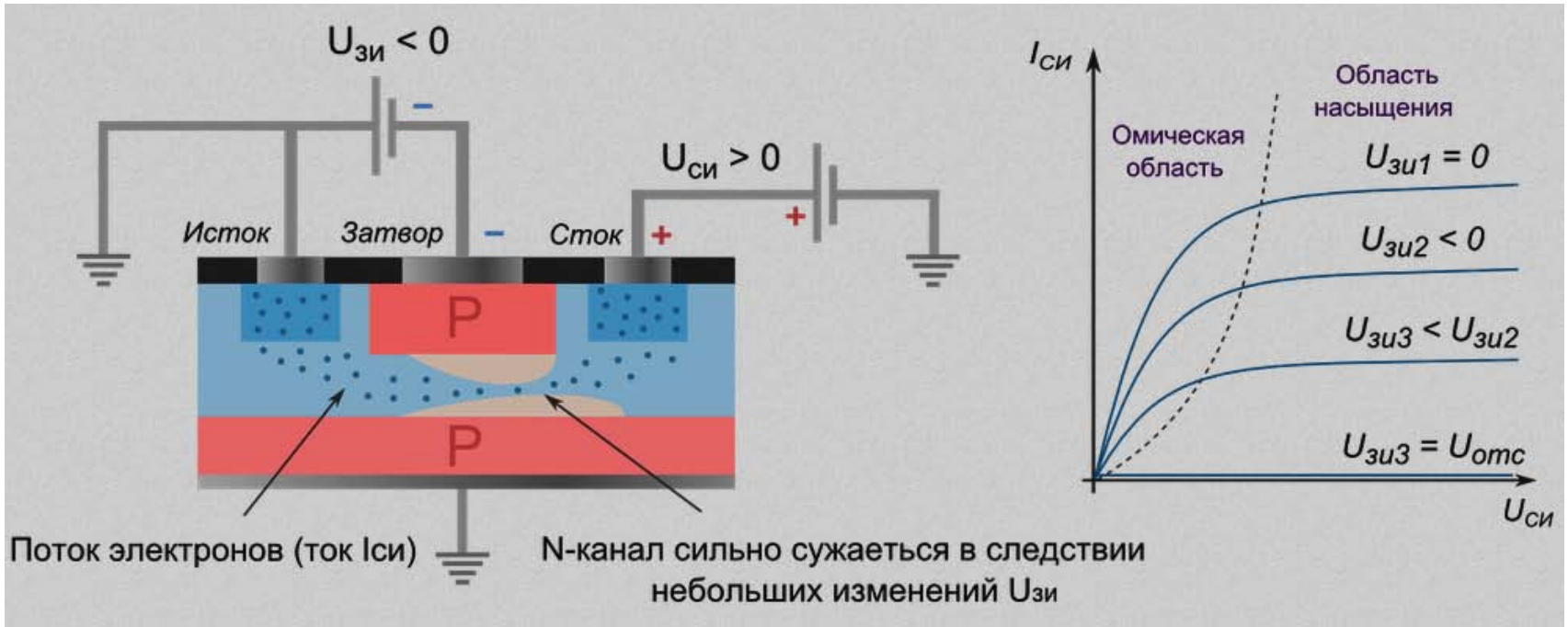
$$K_p = \frac{P_{вых}}{P_{вх}} = \frac{I_к^2 R_к}{I_э^2 R_{вх}} = \frac{\alpha^2 R_к}{R_{вх}} >> 1$$



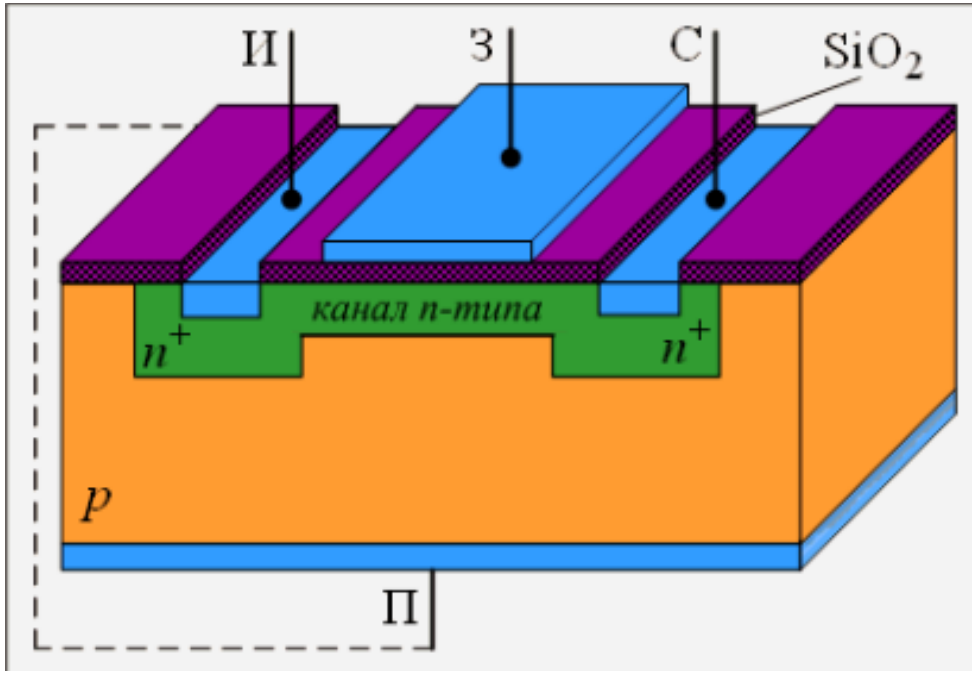
Полевые транзисторы

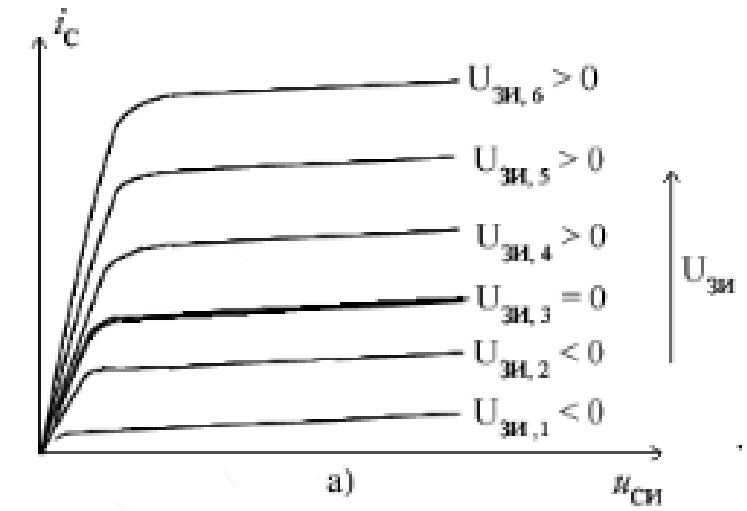


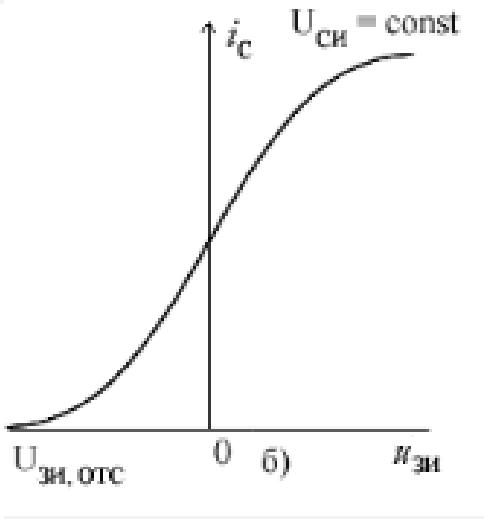




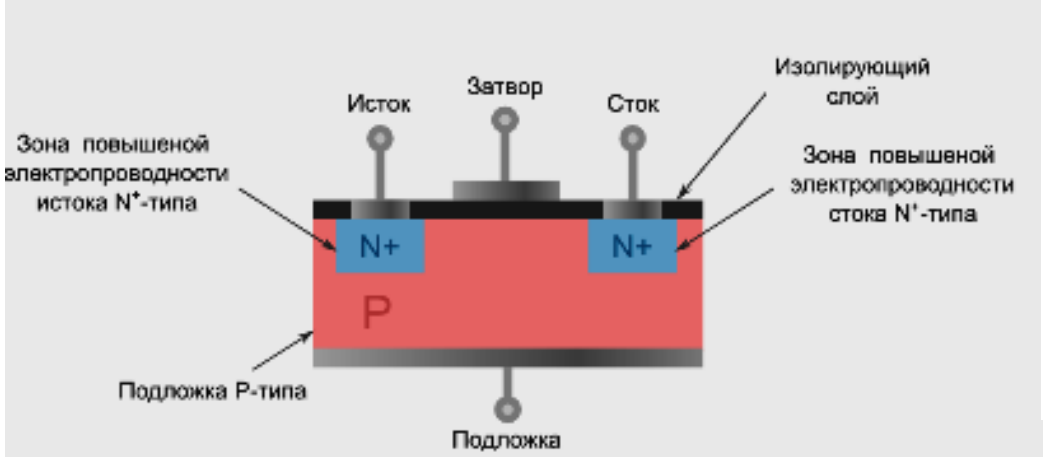
Полевые транзисторы



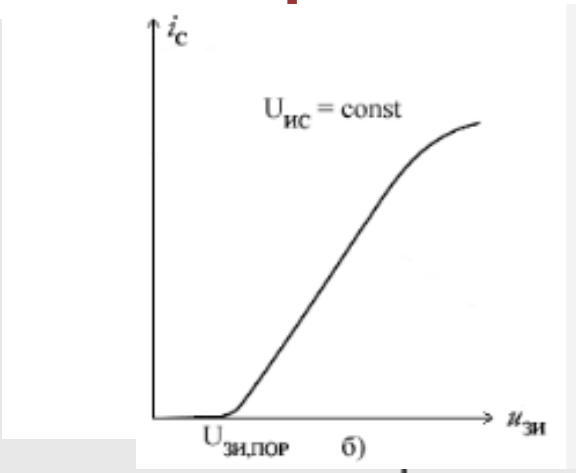


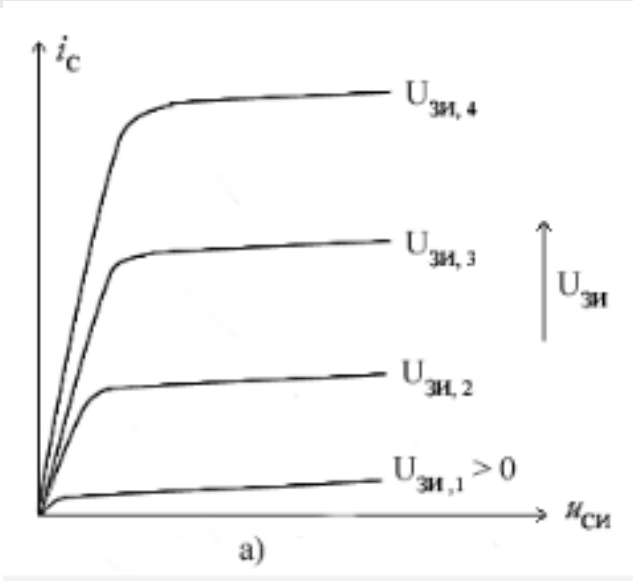


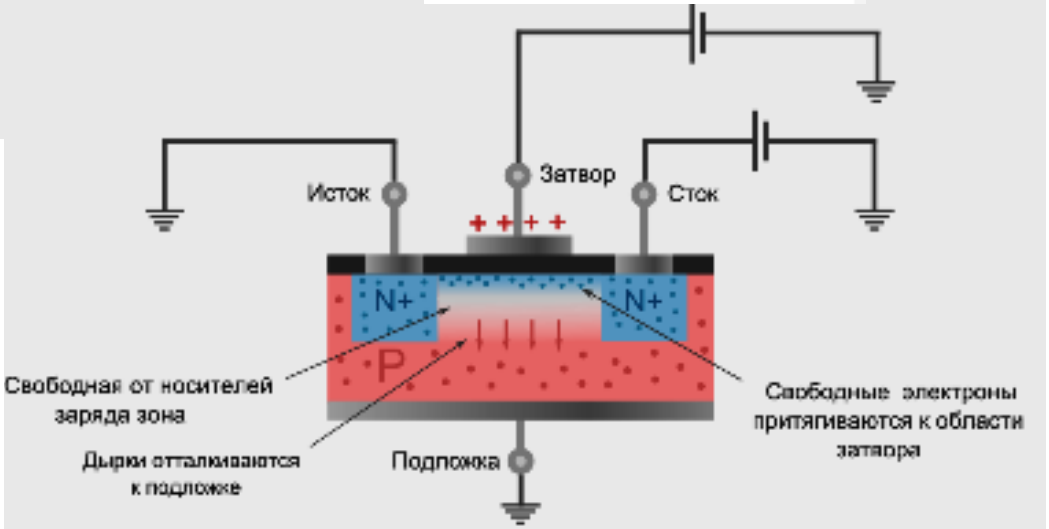
Полевые транзисторы



Устройство МДП-транзистора с индуцированным каналом N-типа



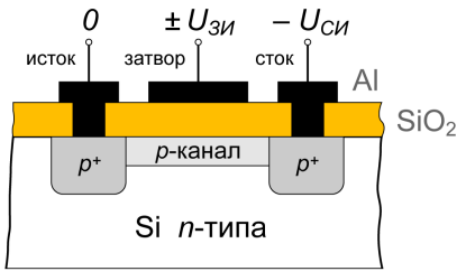


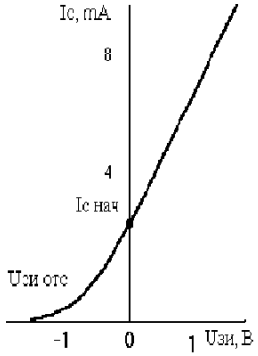


Работа МДП-транзистора с индуцированным каналом N-типа

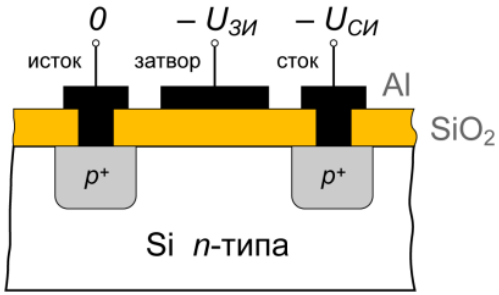
Полевые транзисторы

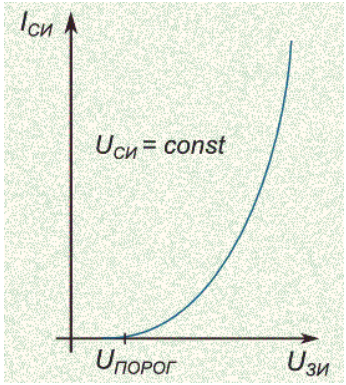
$$g_m = \frac{\partial I_c}{\partial U_з} \cong \frac{\Delta I_c}{\Delta U_з} \text{ при } U_{си} = const.$$

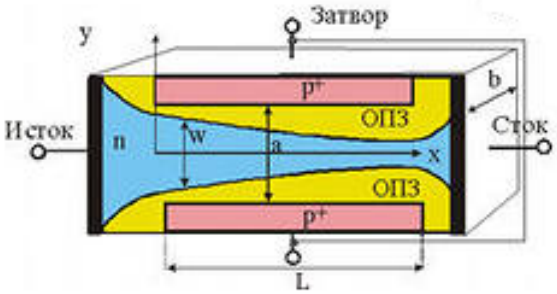


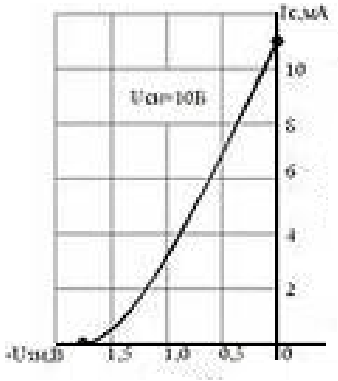


$$S = \left(\frac{\partial I_c}{\partial U_з}\right)_{U_c = const} ; \quad r_c = \left(\frac{\partial U_c}{\partial I_c}\right)_{U_з = const}$$









Полевые транзисторы

С управляющим

p—n-переходом

МОП- или МДП-транзисторы

(полевые транзисторы с изолированным затвором)

n-канал

p-канал

Со встроенным каналом

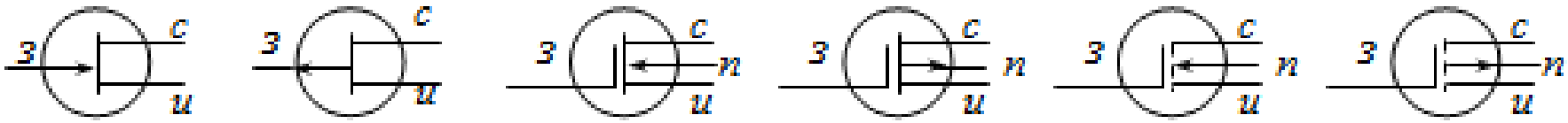
n-канал

p-канал

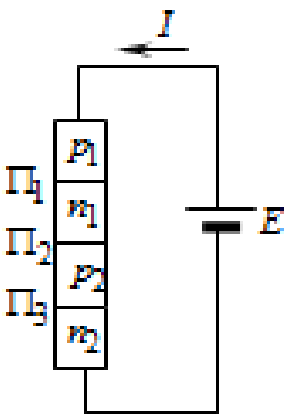
С индуцируемым каналом

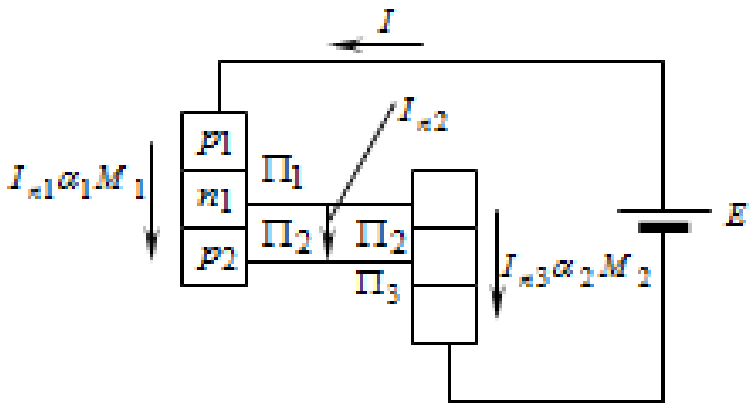
n-канал

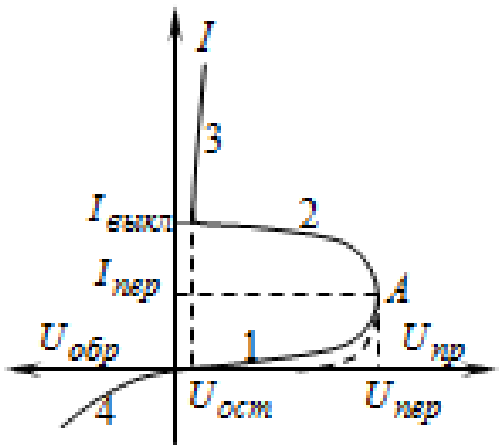
p-канал



Тиристоры







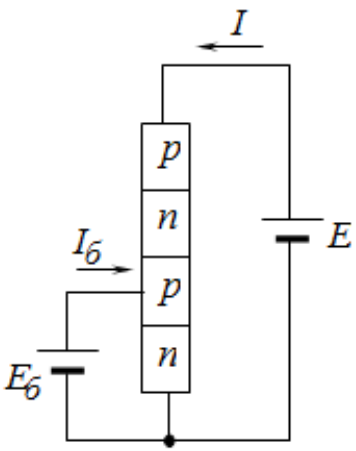
. При $M_1 = M_2 = M$

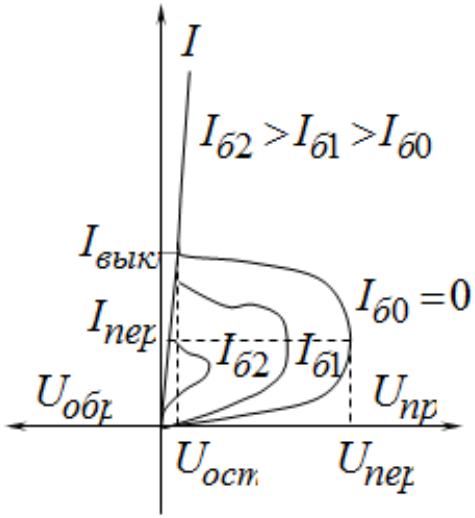
$$I_{П2} = \left(I_{П1}\alpha_1 + I_{П3}\alpha_2\right)M + I_{К0}.$$

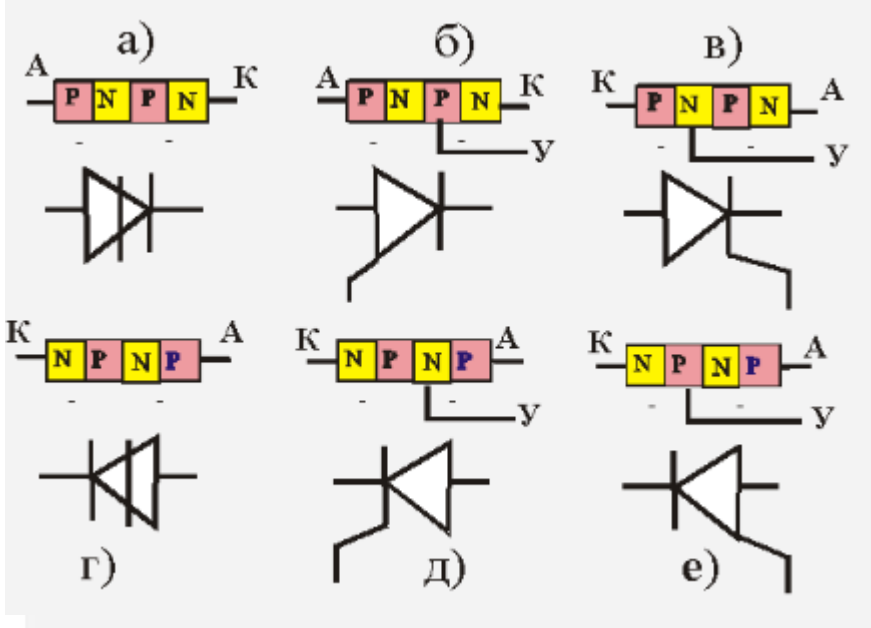
Через все переходы протекает одинаковый ток $I_{П1} = I_{П2} = I_{П3} = I$. Тогда

$$I = \frac{I_{К0}}{1 - \left(\alpha_1 + \alpha_2\right)M}.$$

Тиристоры







тиристоры: динисторы (а,г), управляемые **тиристоры** (б)